

Network Signaling (SS7)

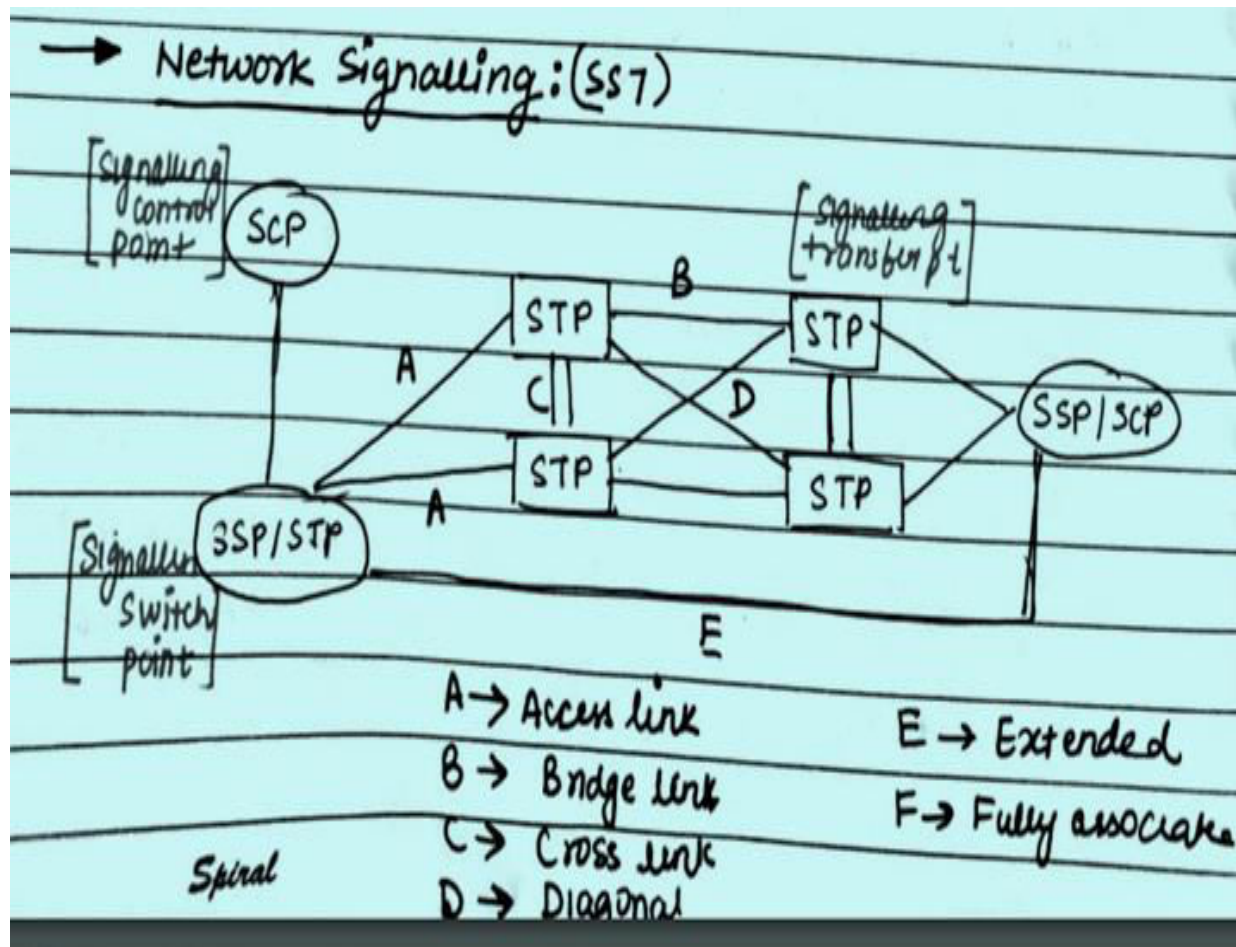
- Introduction
- Architecture
- Performance
- Advantages
- Multiple Access Techniques

By Jasleen Kaur
IT Dept.
GTBIT

Introduction

- The SS7 signaling protocol is widely used for common channel signaling between interconnected networks.
- SS7 is used to interconnect most of the cellular MSC's & is the key factor in enabling autonomous registration and automated roaming in first generation cellular systems.

Network Signaling SS7



- **SSP(Signaling Switching Point)**

- It is an interface to telephone network.
- It convert voice signal to SS7 signal link.
- Local exchange to subscriber.

- **SCP (Signaling Control Point)**

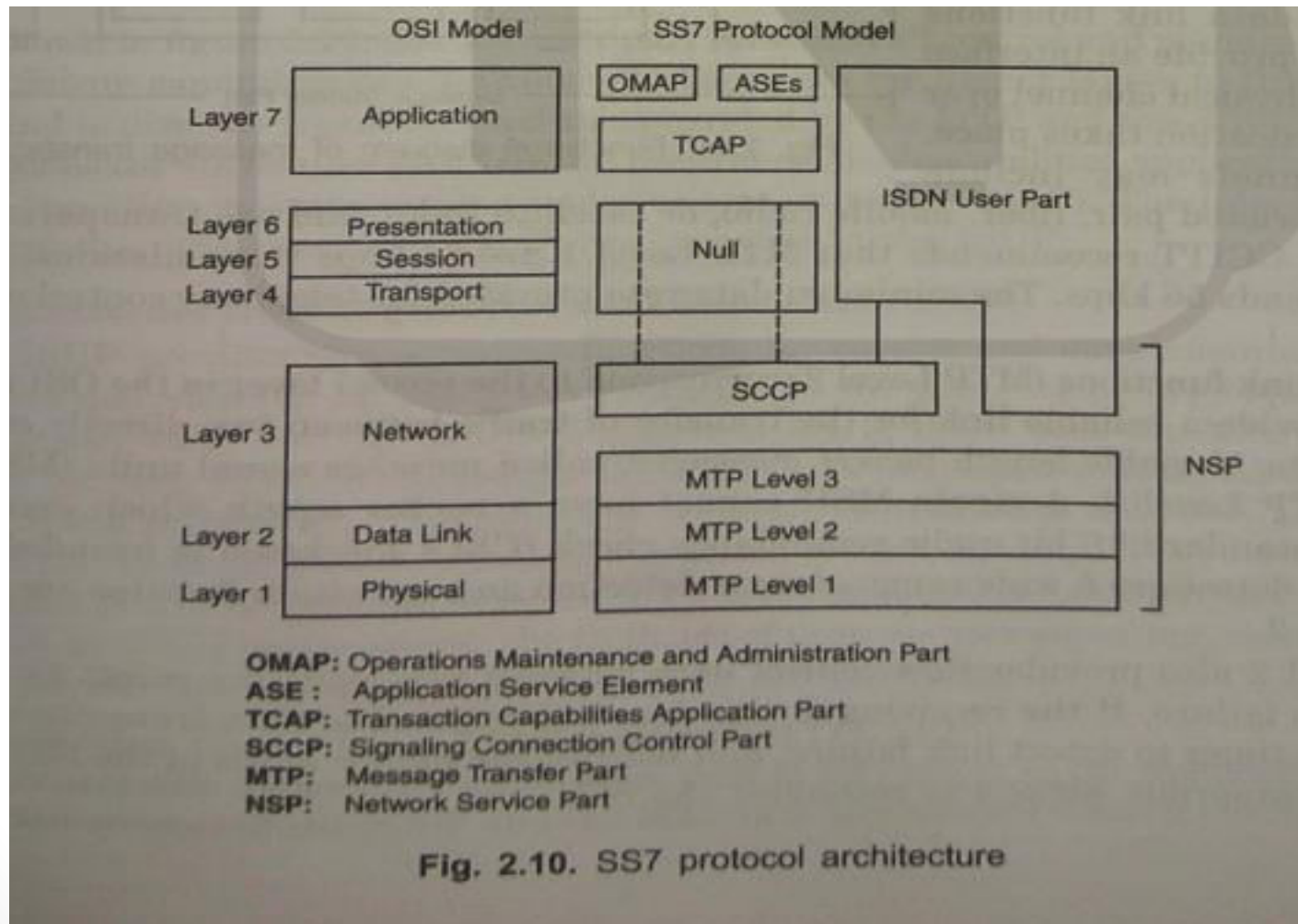
- It is an interface between Database (HLR,VLR)
- Handles database queries & subscriber information.

- **STP(Signaling Transfer Point)**

- It is basically a network node.
- Works as a router.

- A Link: connects SCP/STP to SSP and provides information about subscriber.
- B Link: connects STP to STP.
- C link: connects STP=STP only works for route failure and in that case provide another part.
- D link: connects STP-STP diagonally.

SS7 Protocol Architecture



- The SS7 uses, a four layer protocol stack.
- These protocol provide different services, depending on the use of, the signaling network.
- The Bottom three layers are meant for communication and transmission of message. These three levels are referred to the Message Transfer Part (MTP).
- MTP provides the reliable service for routing messages through the SS7network.
- The upper portion or the fourth layer of the stack, performs the data processing function.

- **Network Service Part (NSP):** provides ISDN nodes with a highly reliable and efficient means of exchanging signaling traffic using connectionless services.
- **Message Transfer Part (MTP):** The MTP provides reliable service for routing messages. The MTP is divided into three levels:
 - **MTP level1:** is the lowest level. MTP1 is the physical layer.
 - It is an interface to actual physical channel.
 - It moves bits and bytes, over electrical and data paths.
 - It uses time slot of a 2Mbps.
 - **MTP level2:** Equivalent to data link layer in OSI.
 - MTP-2 provide support to MTP-3 in link management by ensuring accurate end to end transmission of a message,

- By performing error detection.
- Arranging of retransmission.
- Implementing flow control,.
- By delivering the message in sequence to users.
- Detecting and declaring faulty link conditions.

- **MTP level3:**

- Provides message routing between signaling points in SS7 network.
- It ensures that the message can be delivered between signaling points, across the SS7 network irrespective of whether they are directly connected or not.
- MTP3 reroutes traffic, away from failed links and signaling points. MTP3 also control traffic, when congestion occurs.
- Provides nodes addressing, routing and alternate routing.

- **User Part:** is known as level4 in the layered structure of the signaling system.
 - It includes the message, message coding and the protocol necessary, to support telephony and ISDN.
 - This consists of the processes for handling the services being supported by the signaling system.
 - The message transfer part, is capable of, supporting many different user parts.
 - The SS7 user part includes the ISDN user part (ISUP), the transaction capabilities application part (TCAP), and the operation maintenance and administration part (OMAP). The telephone user part(TUP) and the data user part(DUP) are included in ISUP.

- 1) Integrated Services Digital Network User Part (ISUP): provides signaling functions for carrier and supplementary services for voice, data and video in an ISDN environment.
 - 2) Transaction Capabilities Application Part(TCAP) refers to the application layer which invokes the services of SCCP and the MTP in a hierarchical format and is concerned with remote operations.
 - 3) Operation Maintenance And Administration Part(OMAP): include monitoring, coordination and control function to ensure that trouble free communications are possible.
- Signaling Connection Control Part(SCCP): can address applications, within a signaling point.

- **Performance of SS7:**

The performance of Signaling network is studied by the connection set-up time(response time) or end to end signaling information transfer time.

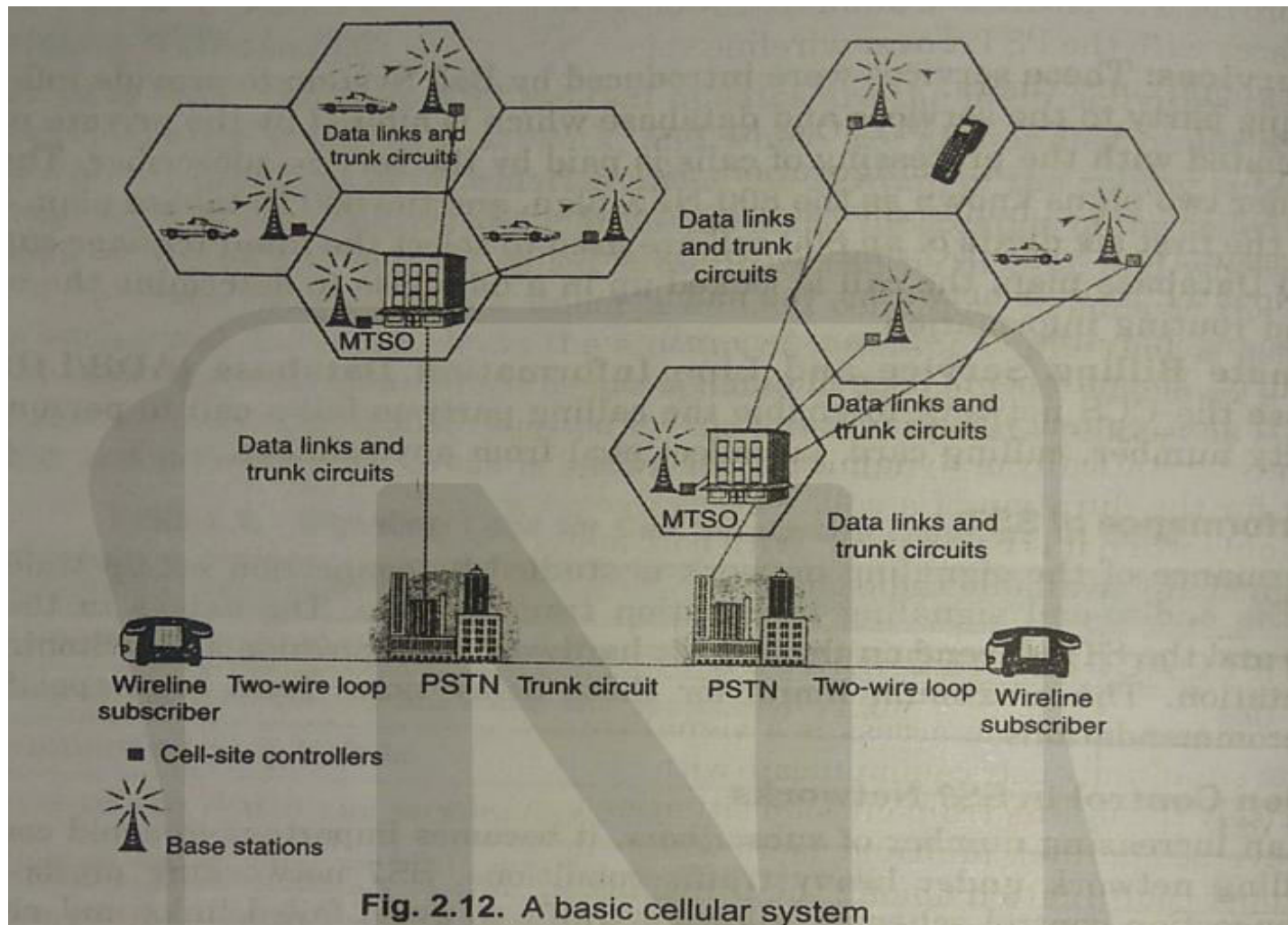
- **Advantages of Common Channel Signaling over Conventional Signaling:**

- 1) Faster call set-up: In CCS, high speed signaling networks are used for transferring the call set-up messages resulting in smaller delay times compared to conventional signaling.
- 2) Greater Trunking (or Queuing) Efficiency: CCS has shorter call set-up and tear down times that results in less call- holding time, reducing the traffic on the network.
- 3) Information Transfer: CCS allows the transfer of additional information along with the signaling traffic providing facilities such as caller identification and voice or data identification.

A Basic Cellular System

- A wireless radio network covering a set of cells in which two way radio units such as cellular or PCS telephones, can communicate.
- The radio network is defines by a set of Base station which serves as central control for all users within that cell.
- Mobile users can communicate directly with the base stations and base station directly with Mobile Telephone Switching Office(MTSO) or MSC.
- An MTSO controls channel assignment, call processing, call setup and call termination which include signaling switching, supervision , and allocating radio- frequency channel.
- MTSO provides a centralized administration and maintenance point for the entire network.

- Base station can improve the transmission quality, but they cannot increase the channel capacity within the fixed bandwidth of the network.
- Base station are distributed over the area of system coverage and are managed and controlled by an on-site computerized cell site controller that handles all cell-site control and switching function.
- The function of Base station is to provide an interface between mobile telephone sets and the MTSO. BS communicate with the MTSO over dedicated data links.



Multiple Access Techniques

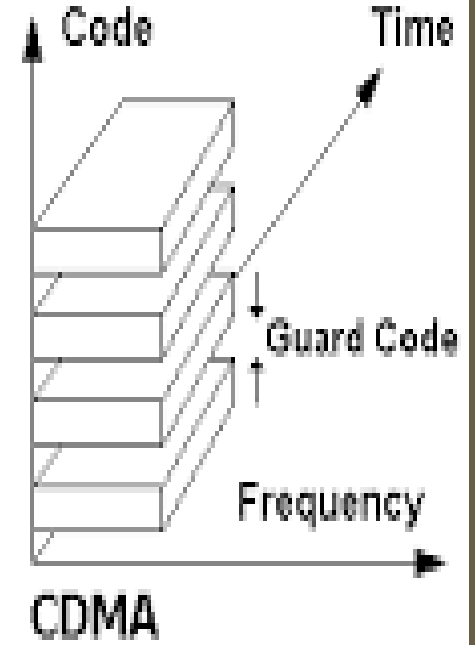
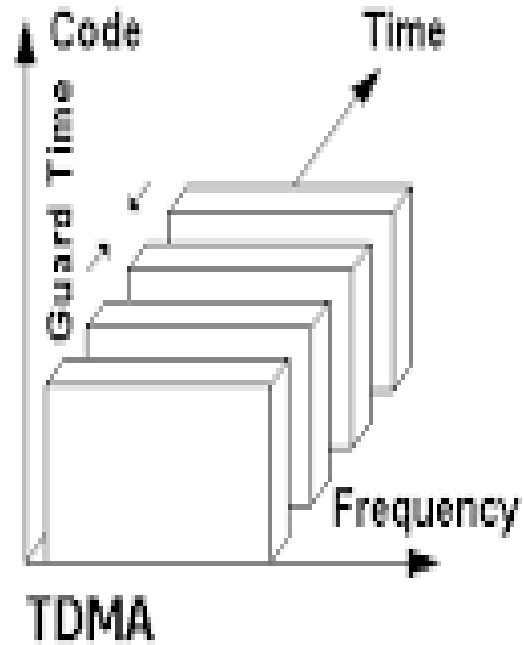
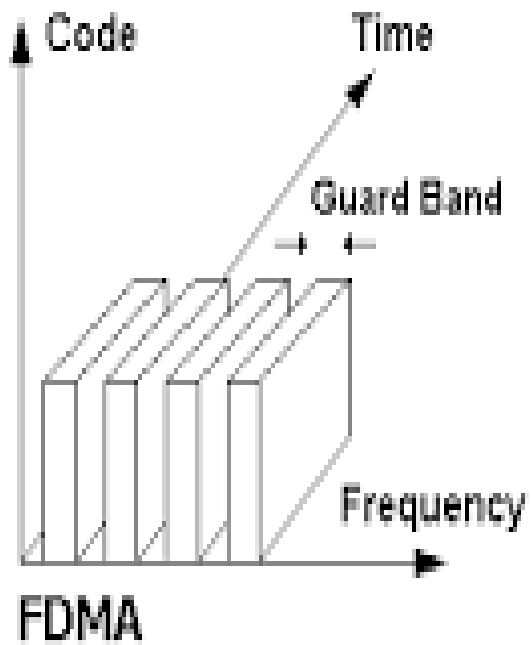
- Multiple access is a signal transmission situation in which two or more users wish to simultaneously communicate with each other using the same propagation channel.
- It is often desirable to allow subscriber to send information simultaneously from MS to BS while receiving information from BS to MS.
- The main aim is able to increase the capacity of channel as possible in given bandwidth with sufficient quality of service.
- Multiple Access Schemes are:
 - 1) Frequency Division Multiple Access(FDMA)
 - 2) Time Division Multiple Access(TDMA)
 - 3) Code Division Multiple Access(CDMA)
 - 4) Space Division Multiple Access(SDMA)

1) FDMA :

- Individual user are allotted individual channels. The channels are frequency band is unique for each user/subscriber. Or
- In this scheme the avail bandwidth is divided into various frequency band.
- Each station allocated a band to send its data when service request is made.
- Thus each band is reserved for specific station and no other user can share the same frequency band.
- Guard Band; In each band there is a guard band which prevents the overlapping or we can say to avoid frequency overlapping small gap in the frequency band are required, those are called guard band.
- Used for FDD (Frequency Division Duplexing).
- Bidirectional communication is possible simultaneously and is essential for cellular communication also. Such a simultaneously talk and listen facility is known as Duplexing. Under frequency division multiplexing, duplexing is done using frequency & termed as FDD.

- **Salient feature of FDMA:**

- i. FDMA requires proper filtering at receiver side to avoid adjacent channel interference.
- ii. In FDMA scheme, if a channel is not in use it will be idle and it will not be used by some other user.
- iii. FDMA channel can handle one phone circuit at a time.
- iv. Complexity is less.
- v. If voice channel are assigned in FDMA , then, the mobile unit and base station start transmitting simultaneously.



2) TDMA :

- In time division duplexing (TDD) utilizes time instead of frequency.
- Each TDD channel has individual time slots for forward and reverse time slots to have bidirectional communication.
- All sender use same frequency at different point of time.

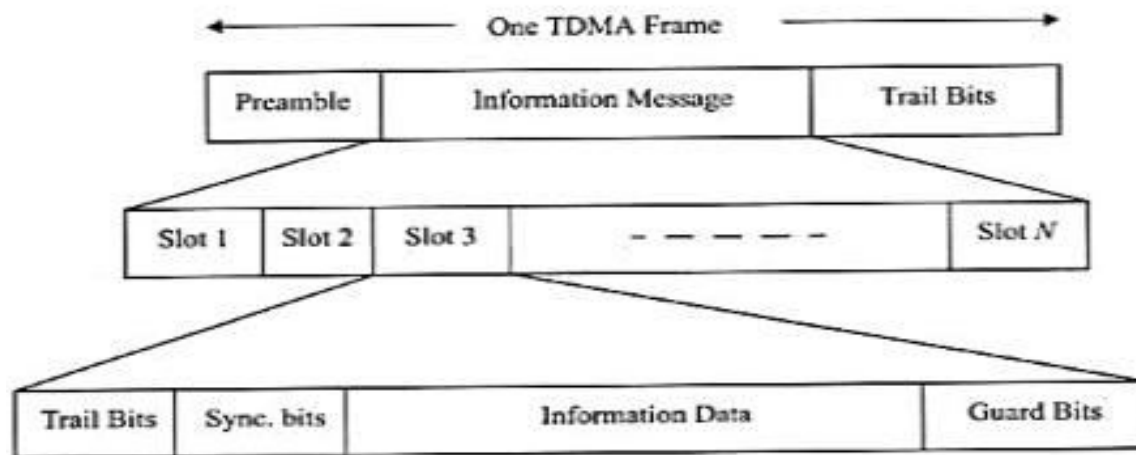


Fig: TDMA frame structure

- In TDMA system the entire radio spectrum is divided into time slots where a single user is allowed to use the radio channel in a time slot.
- In this format the preamble field contains the synchronization and address information. The guard bits are used to allow synchronization of different receiver between various frames and time slots. There are 'N' number of time slots for 'N' number of users.
- **Salient feature of TDMA:**
 - i. Each user in TDMA multiple access scheme shares same carrier frequency but with non overlapping time slots.

3) CDMA:

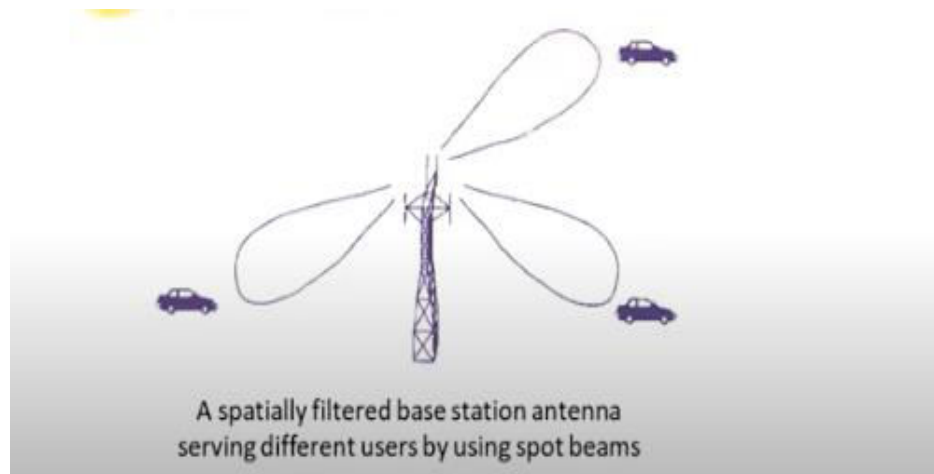
- In CDMA technique any users share the same carrier frequency.
- The narrow band signal is also multiplexed with a spreading signal of larger bandwidth. This spreading signal is actually a pseudo noise code sequence.
- Spread Spectrum technology ,as it allow many users to occupy the same time and frequency in a given band/space.
- Each user /subscriber is allocated n individual pseudo random codeword that is orthogonal all other codewords of remaining users.

- **Salient feature of CDMA:**

- i. CDMA has a better soft capacity limit than TDMA and FDMA.
- ii. CDMA has less interference problems due to allocation of unique codes to each user and multiple access technique is applied for defence areas than other multiple access technique.
- iii. Radio signal strength indicator(RSSI) is being used in CDMA to have better power control.
- iv. Handoff is very well handled.
- v. Recommended for voice and data communication.
- vi. Every user use full available spectrum.

- **SDMA:**

- SDMA is capable of controlling radiated energy for each and every user in space, i.e. by spot beams a spatially filtered base station antenna serves different cellular users.
- SDMA is used for allocating a separated space to users in wireless network.



- SDMA serves different users by using spot beam antennas.
- A typical application involves assigning an optimal base station to a mobile phone user.
- Typically SDMA is never used in isolation but always in combination with one or more schemes such as FDMA, TDMA or CDMA.
- **Problems with SDMA:**
 - i. Reverse link presents the difficulty in cellular system.
 - ii. Different propagation path exists from user to the base station.

Comparison

SDMA/TDMA/FDMA/CDMA

Approach	SDMA	TDMA	FDMA	CDMA
Idea	segment space into cells/sectors	segment sending time into disjoint time-slots, demand driven or fixed patterns	segment the frequency band into disjoint sub-bands	spread the spectrum using orthogonal codes
Terminals	only one terminal can be active in one cell/one sector	all terminals are active for short periods of time on the same frequency	every terminal has its own frequency, uninterrupted	all terminals can be active at the same place at the same moment, uninterrupted
Signal separation	cell structure, directed antennas	synchronization in the time domain	filtering in the frequency domain	code plus special receivers
Advantages	very simple, increases capacity per km ²	established, fully digital, flexible	simple, established, robust	flexible, less frequency planning needed, soft handover
Dis-advantages	inflexible, antennas typically fixed	guard space needed (multipath propagation), synchronization difficult	inflexible, frequencies are a scarce resource	complex receivers, needs more complicated power control for senders
Comment	only in combination with TDMA, FDMA or CDMA useful	standard in fixed networks, together with FDMA/SDMA used in many mobile networks	typically combined with TDMA (frequency hopping patterns) and SDMA (frequency reuse)	still faces some problems, higher complexity, lowered expectations; will be integrated with TDMA/FDMA

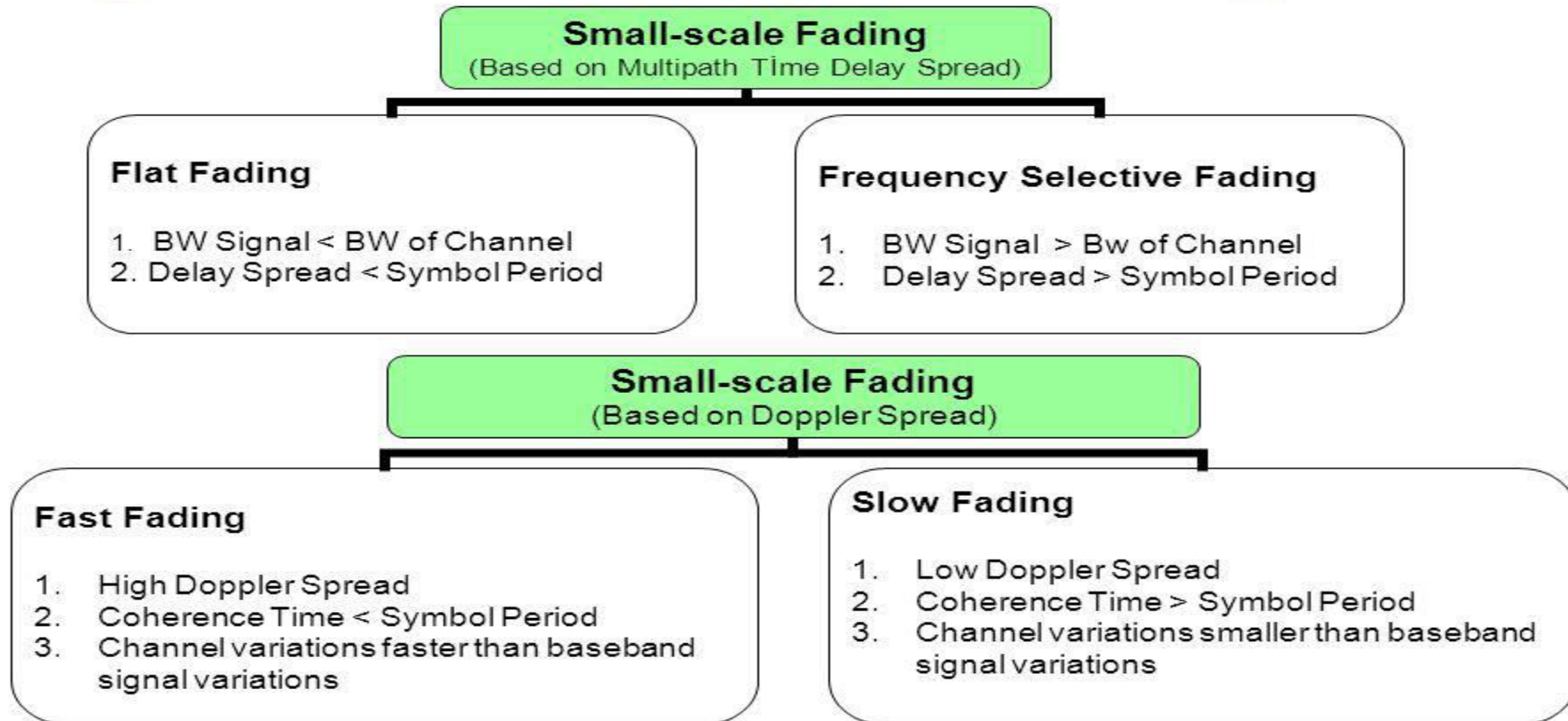
SMALL SCALE FADING

- This refers to the fluctuations in signal strength and phase over short distance and small duration of time. It is also called **Rayleigh Fading**. Small Scale Fading affects almost all forms of wireless communication and overcoming them is a necessity to increase efficiency and decrease error.
- Reasons of small scale fading:
 - i. Rapid change in signal strength over small distance
 - ii. Time dispersion due to multipath delays

TYPES OF SMALL SCALE FADING

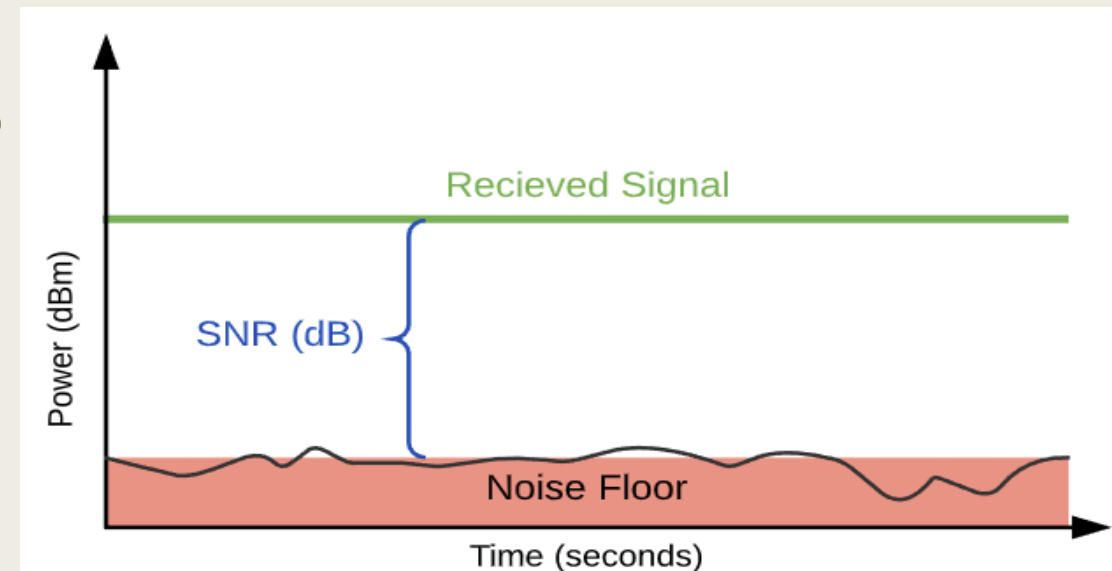
- Types of fading experienced by Tx signal depends on the nature of Tx signal with respect to characteristics of channel

Types of Small-scale Fading



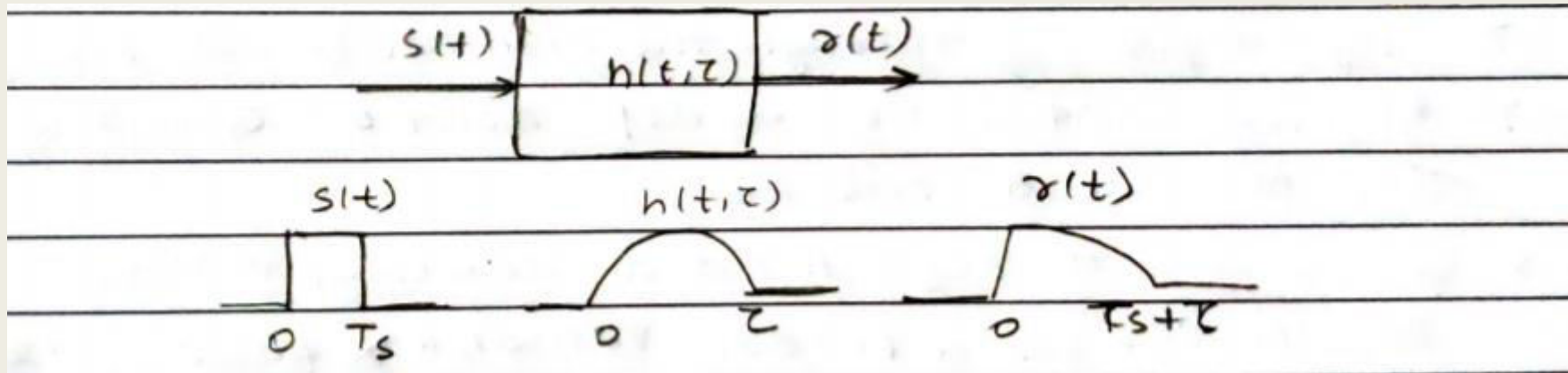
- **Fast Fading:** It occurs mainly due to reflections for surfaces and movement of transmitter or receiver.
- High doppler spread(*spectral broadening*) is observed in the fast fading with Doppler bandwidth comparable to or greater than the bandwidth of the signal and the channel variations are as fast or faster than the signal variations
- It causes linear distortions in the shape of the baseband signal and creates Inter Symbol Interference (ISI). One way to remove ISI is adaptive equalization(It is an equalizer that automatically adapts to time-varying properties of the communication channel)
- Rate of change of channel is faster than rate of change of Tx signal
- Doppler spread is greater than Tx signal
 - i. $T_s > T_{\text{channel}}$ (Ts- Symbol Period)
 - ii. $B_{\text{signal}} < B_{\text{doppler spread}}$

- **Slow Fading:** It occurs mainly due to shadowing where large buildings or geographical structures obstruct the LOS (Line of sight (LOS) is the level of obstruction on the path between two points).
- Low doppler spread is observed in Slow Fading with the doppler bandwidth being smaller compared to the bandwidth of the signal and the channel variations are slow relative to the signal variations.
- It results in reduction of SNR (**Signal-to-Noise Ratio (SNR)** is the difference between the received wireless signal and the noise floor) which can be overcome using error correction techniques and receiver diversity techniques.
- Rate of change of channel is slower than Tx signal
- Impulse response is static
- Doppler spread is smaller than Tx signal
- i. $T_s \ll T_c$ (Tc- Coherence Time)
- ii. $B_{\text{signal}} \gg B_{\text{doppler}}$ (Ts- Symbol Period)



- **Multipath Fading:** It occurs when a signal reaches the receiver from various path i.e. when multipath propagation takes place.
 - Multipath fading can affect all ranges of frequencies starting from low frequency to microwave and beyond.
 - It affects both the amplitude and the phase of the signal causing phase distortions and ISI.
- Multipath fading can affect signal transmission in two ways:
 - **Flat Fading:** In flat fading, all frequency components get affected almost equally. Flat multipath fading causes the amplitude to fluctuate over a period of time.
 - Occurs in rx signal if channel has constant gain and linear phase response over $B_{\text{channel}} > B_{\text{transmitted signal}}$
 - **Frequency Selective Fading:** refers to multipath fading when the selected frequency component of the signal is affected.
 - It means selected frequencies will have increased error and attenuation as compared to other frequency components of the same signal.
 - Caused due to time dispersion due to delayed multipath components
 - Occurs in rx signal if channel has constant gain and linear phased response over $B_{\text{channel}} < B_{\text{signal}}$

- Received signal includes faded and delayed signal hence signal is distorted and Intersymbol Interferences is induced



Wireless Channel and Diversity

- Introduction
 - Fading
- Types of fading
 - Diversity
- Types of Diversity

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IT dept.
GTBIT

- Wireless Communication is prone to noise, interference and other channel obstruction which change over time unpredictably resulting degradation of signal. Losses in signal arises both in transmission of signal in short distances well as long distance.

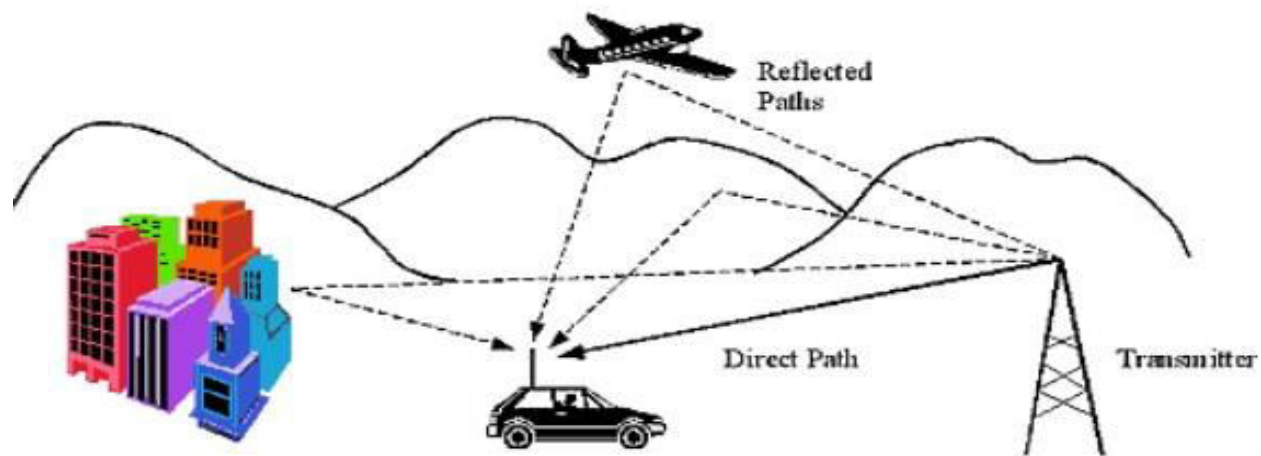
- **Two types of communication channels:**

- 1) Wired channels
- 2) Radio or wireless channels.

The wired channels are stationary and predictable whereas the radio channels are extremely random and do not offer easy analysis. Even the speed of motion affects how rapidly the signal level fades as a mobile terminal moves in space.

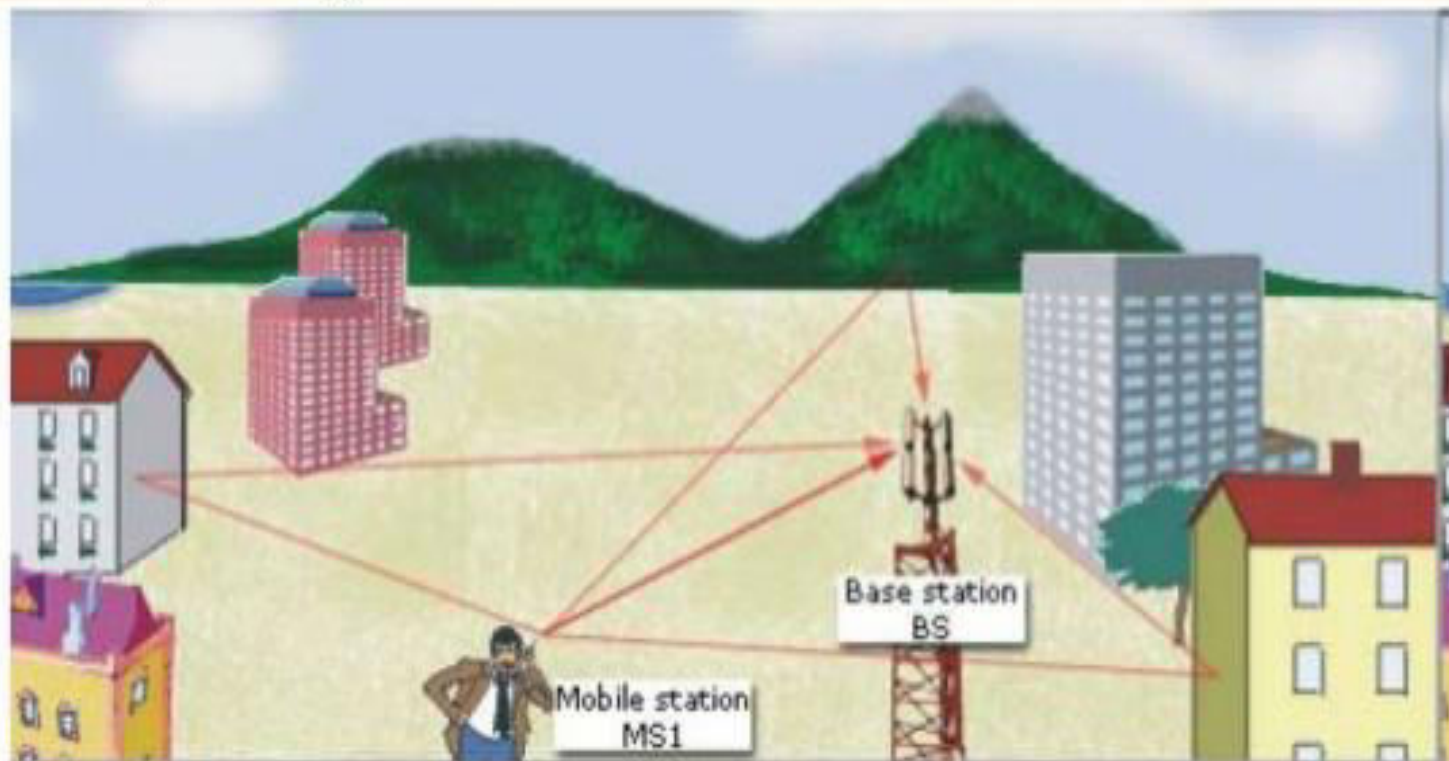
- The signal can be either reflected, diffracted or scattered by obstacles (i.e. buildings, vehicles, trees, mountains etc.) so that multiple copies through multiple paths of the signal with varying delays can be received is known as multipath propagation. The three basic propagation mechanism (i.e. reflection, diffraction & scattering) mainly responsible for multipath propagation.
- Multipath propagation mainly produce three effects in wireless communication:
 - i. Multipath copies of signal may arrive at different phases.
 - ii. Multipath propagation produce Inter Symbol Interference(ISI).
 - iii. Multipath propagation produce short term or small scale fading.

- **Fading:** The strength(power) of received signal varies with respect to time is known as fading.
- The main causes of fading are attenuation (loss of signal strength), changes in transmission medium, refraction , multipath propagation, rainfall, obstacles etc.
- Fading can be divided into two times:
 - 1) Large Scale (long term) fading
 - 2) Small Scale (short term) fading



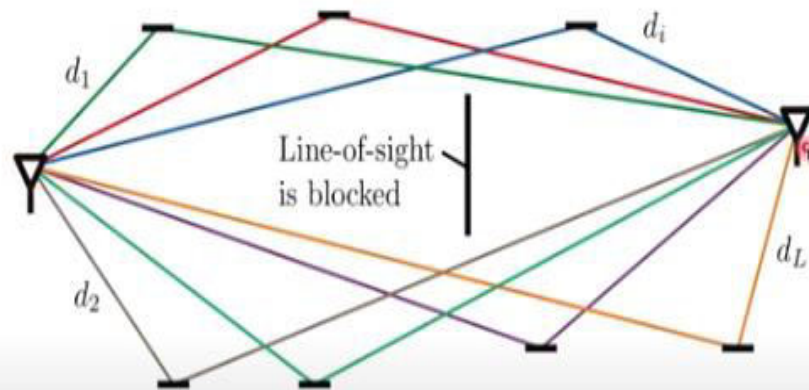
How Fading Occur ????

- *Fading* signals occur due to reflections from ground & surrounding buildings (clutter) as well as scattered signals from trees, people, towers, in a Large Area.



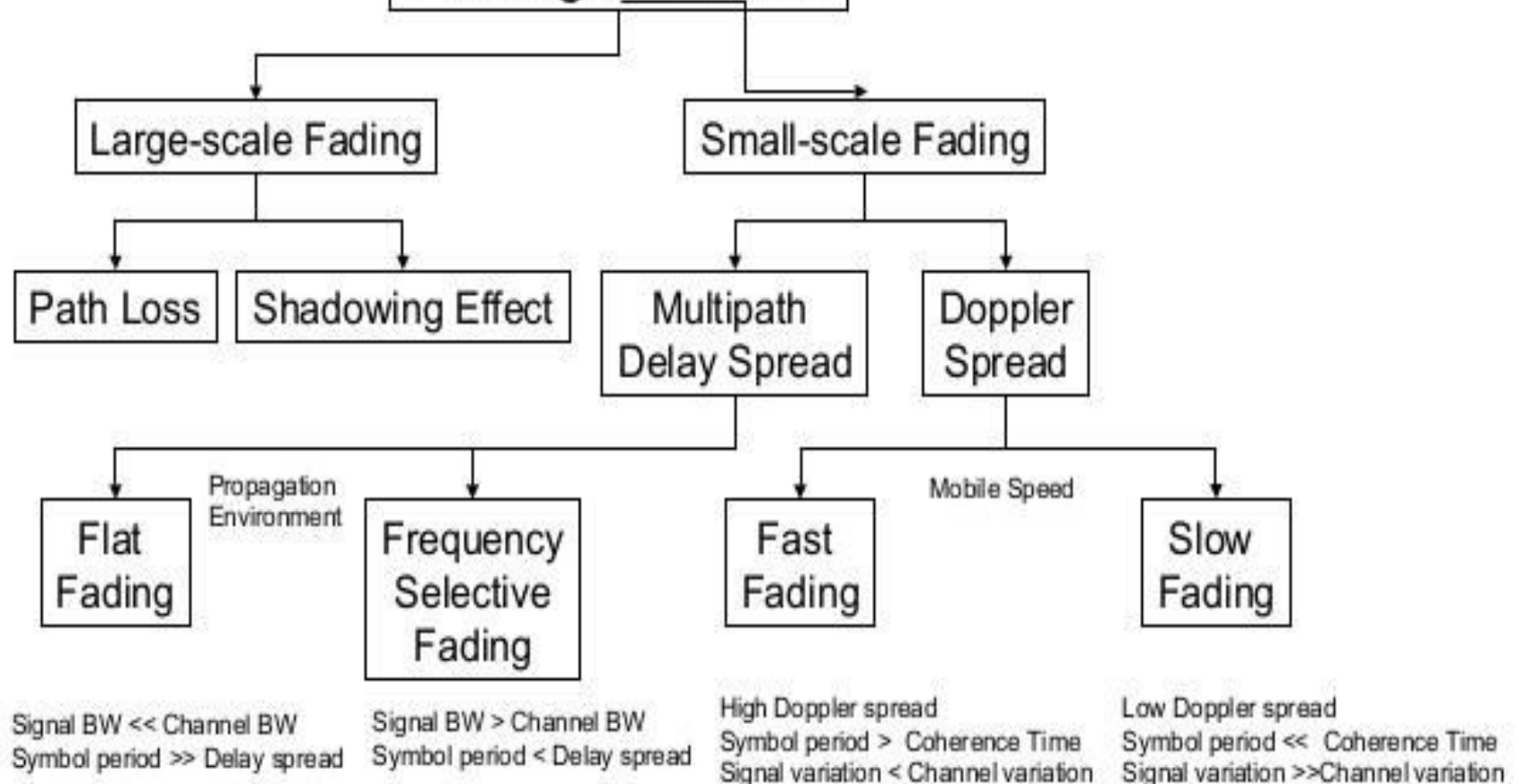
- Fading is the most important thing when you design any RF(radio frequency) Design.
- The communication between the base station and mobile station in mobile system is mostly non-LOS (line of sight)
- The LOS path between transmitter and the receiver is affected by terrain(natural terrain; mountains) and obstructed by buildings and other objects.
- The mobile station is also moving in different direction at different speed.
- The RF signal from the transmitter is scattered by reflection and diffraction and reaches the receiver through many non-LOS paths.
- This non-LOS path causes long term and short term fluctuation in the form of log normal fading and rayleigh and rician fading, which degrades the performance of RF channel.

Multipath propagation



Non-line-of-sight
channel:
• Scattering

Fading Channel



1) Large Scale Fading:

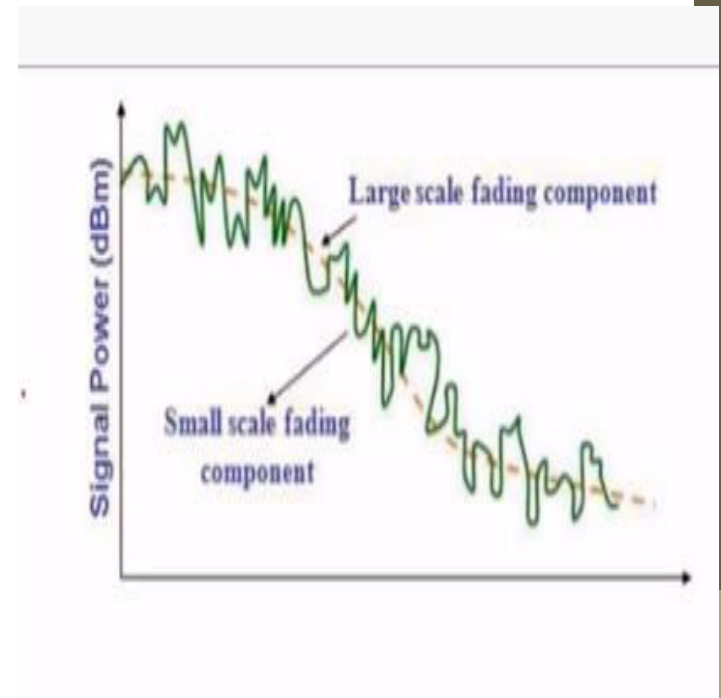
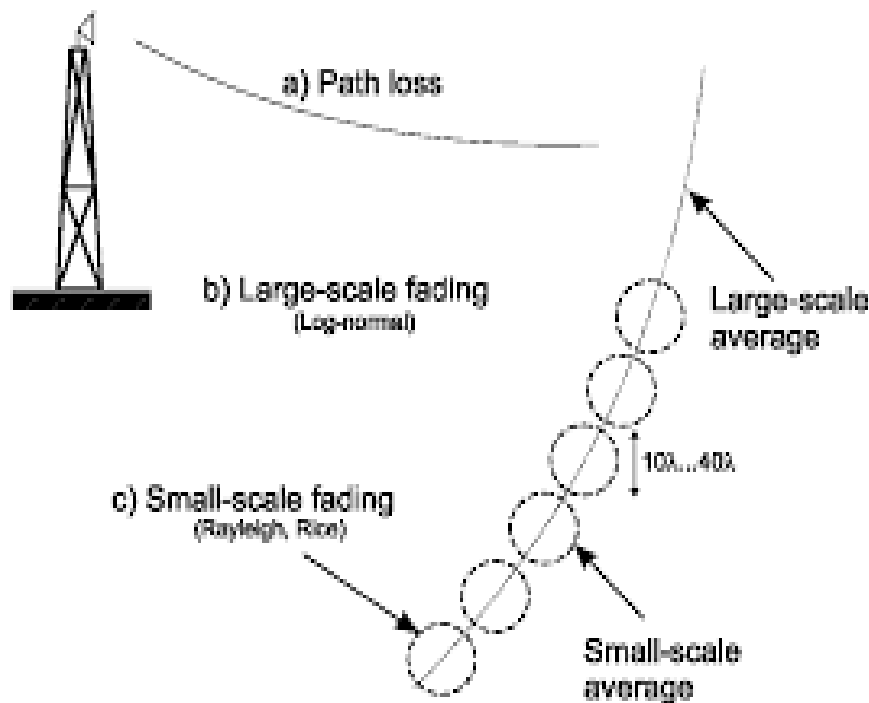
- The local mean (average) power over time duration is known as long term fading. Fading duration is very long as well as signal attenuation(effect) is large due to path loss and shadowing.
- Long term fading is sometimes called 'shadowing'. It changes over relatively long periods often due to more distant objects.
- Large scale fading occurs when an obstacle gets positioned between the wireless device and the signal transmitter. The interference causes significant reduction in signal strength because the wave is shadowed or blocked by the obstacles.
- Path Loss : Degradation of signal due to dissipation in path because of propagation effects.
- Shadowing: Caused by obstacles that absorbs power.

2) Small Scale Fading:

- This refers to the fluctuations in signal strength and phase over short distance and small duration of time. It is also called **Rayleigh Fading**. Small Scale Fading affects almost all forms of wireless communication and overcoming them is a necessity to increase efficiency and decrease error.
- Reasons of small scale fading:
 - i. Rapid change in signal strength over small distance.
 - ii. Time dispersion due to multipath delays.
 - iii. Here loss is caused due to multipath delays and Doppler's spreading.

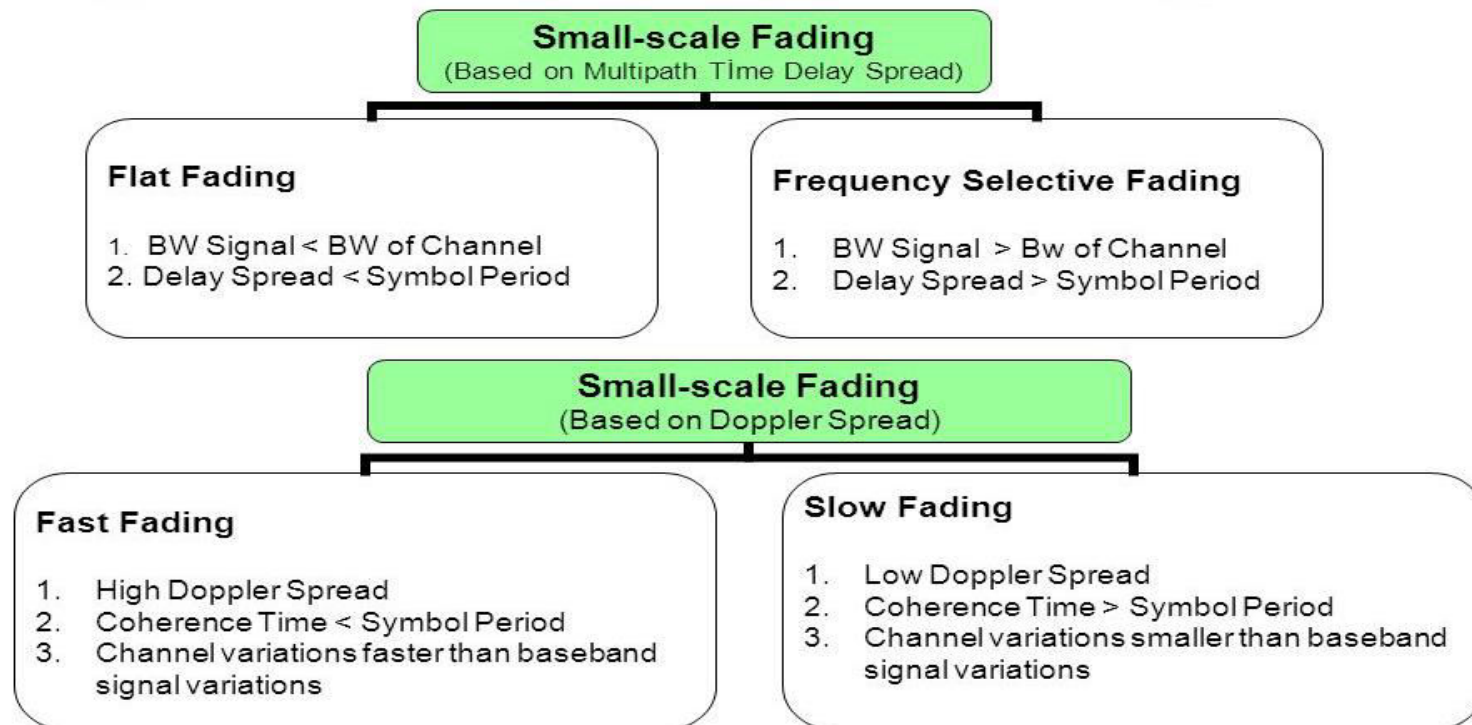
When mobile it is worse as the multi-path effects are continuously changing. Also get Doppler effects – the frequency changes due to movement of objects in the environment including the user.

- Multipath Propagation: Addition of constructive and destructive signal components while propagating.
- Doppler's spreading: Change in frequency due to relative motion of RX with respect to TX.



- Types of fading experienced by Tx signal depends on the nature of Tx signal with respect to characteristics of channel.

| Types of Small-scale Fading



Based on Doppler Spread:

1. Fast Fading Channel:

- i. High doppler spread is observed in the fast fading with Doppler bandwidth comparable to or greater than the bandwidth of the signal and the channel variations are as fast or faster than the signal variations
- ii. It is caused due to frequency dispersion due to Doppler's spreading.
- iii. In Fast fading, the rate of change of channel is faster than rate of change of Tx signal.
- iv. $T_c < T_s$ (Coherence time is less than symbol time)
- v. Doppler spread is greater than Transmitted (Tx) signal Bandwidth.
- vi. $B_{\text{signal}} < B_{\text{doppler spread}}$.

2. Slow Fading:

- i. Low Doppler spread is observed in Slow Fading with the Doppler bandwidth being smaller compared to the bandwidth of the signal and the channel variations are slow relative to the signal variations.
- ii. It is caused due to frequency dispersion due to Doppler spreading.
- iii. In slow fading, the rate of change of channel is slower than rate of change of TX signal.
- iv. $T_s \ll T_c$
- v. Doppler Spread is smaller than transmitted signal (Tx) bandwidth
- vi. $B_{\text{signal}} \gg B_{\text{doppler}}$

- **Based on Multipath Propagation:**

- 1. Flat Fading:**

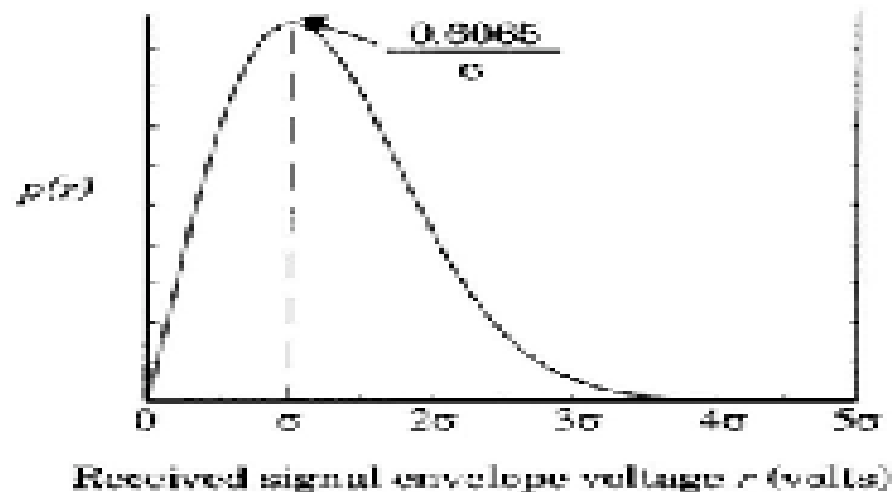
- i. In flat fading, all frequency components get affected almost equally. Flat multipath fading causes the amplitude to fluctuate over a period of time.
- ii. It is caused by time dispersion due to multipath components.
- iii. Occurs in rx signal if channel has constant gain and linear phase response over $B_{\text{channel}} > B_{\text{transmitted signal}}$.

2. Frequency Selective Fading:

- i. FSF refers to multipath fading when the selected frequency component of the signal is affected.
- ii. It means selected frequencies will have increased error and attenuation as compared to other frequency components of the same signal.
- iii. Caused due to time dispersion due to delayed multipath components.
- iv. Occurs in rx signal if channel has constant gain and linear phased response over $B_{\text{channel}} < B_{\text{signal}}$

- **Rayleigh & Rician Fading Channels:** These fadings occur to a signal as a result of effect of propagation channel.
- **Rayleigh Fading Channel:**
 - The phenomenon is due to multipath propagation of the signal.
 - All the signals are NLOS signals and there is no dominant direct path.
 - Signals from all the paths have comparable signal strengths.
 - The instantaneous received power seen by a moving antenna becomes a random variable depending on the location of the antenna.
 - It is assumed that variation in signal passed through Rayleigh fading channel would follow Rayleigh distribution.

- vi. Signal passing through this channel follows Rayleigh distribution.
- vii. It is caused due to reflected and scattered waves.
- viii. When there is no LOS (line of sight) path available between transmitter and receiver, received signal obey Rayleigh Distribution. It is assumed that the magnitude of a signal that has passed through a Tx (transmitter medium) will vary randomly or fade according to Rayleigh Distribution.
- Graphical representation of Rayleigh Probability Density function (P.D.F). :



Rayleigh Fading

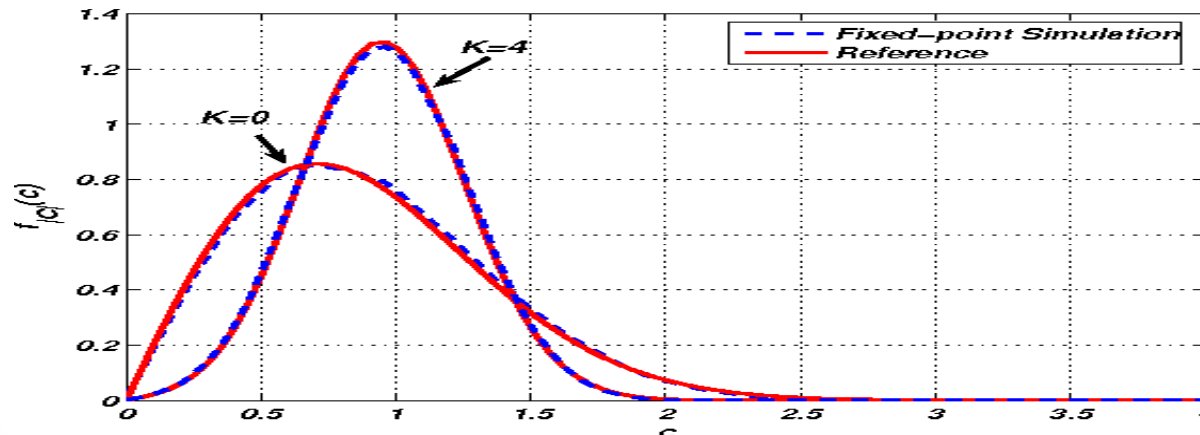
□ The Rayleigh pdf is

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & \text{for } r > 0 \\ 0 & \text{otherwise} \end{cases}$$

Where r is the envelope amplitude of Rx signal & $2\sigma^2$ is the power of the signal

➤ Rician Fading Channel:

- This phenomenon is due to multipath propagation of the signal.
- In this case there is partially scattered field.
- One dominant signal, others are weaker.
- When direct LOS path is available between transmitter and receiver, the received signal obey Rician Distribution.



Rician fading

➤ If number of such stochastic component large & none are dominant
The variables $x_r(t)$ and $y_r(t)$ resulting from their addition.

$$r_0(t) = \sqrt{x_r^2(t) + y_r^2(t)}$$

The received signal = Multiple reflective rays + Significant line-of-sight (non-faded) component, Then :

The received envelope amplitude has a Rician pdf

The fading is referred to as *Rician fading*

$$p(r_0) = \begin{cases} \frac{r_0}{\sigma^2} \exp \left[-\frac{(r_0^2 + A^2)}{2\sigma^2} \right] I_0 \left(\frac{r_0 A}{\sigma^2} \right) & \text{for } r_0 \geq 0, A \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

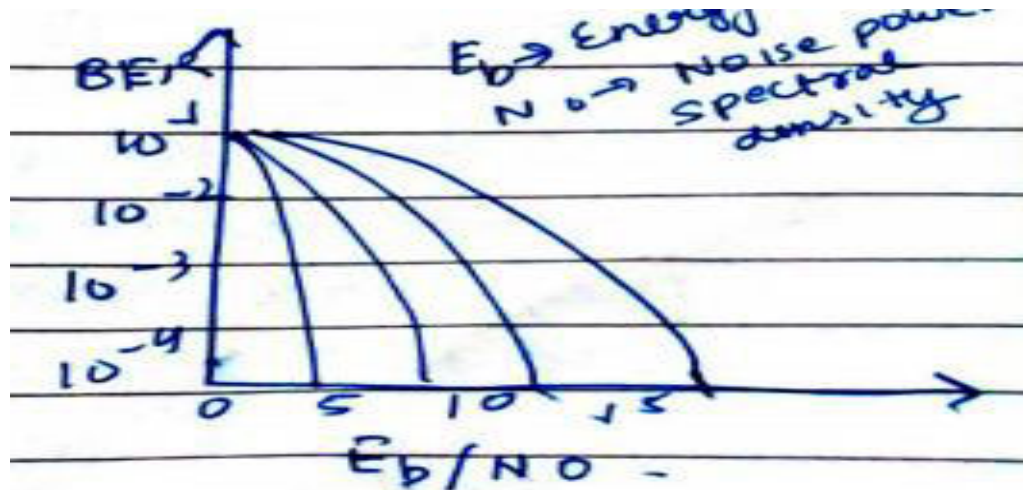
A : the peak amplitude of the non fading signal component called (*specular component*)

σ^2 : the predetection mean power of the multipath signal.

I_0 : the modified Bessel function of the first kind and zero order.

BER performance in fading Channels:

- BER stands for Bit Error Rate per bit transferred.
- BER is a function of E_b/N_0 and is plotted for various fading channel.
- E_b is average energy per bit, N_0 is noise power density.
- As BER decreases, E_b/N_0 increases



- **Conclusion:**

- i. AWGN exhibit good performance. (Additive White Gaussian Noise channel)
- ii. Rician fading good performance.
- iii. Rayleigh fading bad performance.
- iv. Flat fading poor performance.

Diversity Modeling

- This was introduced to compare and overcome deep fading.
- As in deep fading if we have single channel the connection will get interrupted or link will failed due to fading. Hence, to remove this problem.
- Multiple links are applied between transmitter and receiver which not affect communication even after fading.
- Diversity is divided into two parts:
 - a) Microscopic Diversity
 - b) Macroscopic Diversity

a) Microscopic:

- Small signal fading can be minimized by this technique.
- It can prevent small scale fading by using two antennas separated by small distance.
- By selecting strong signal at all time a receiver can mitigate small scale fading.

b) Macroscopic:

- In large scale fading , the signal strength reduces because of shadowing process.
- In this Base station separated by long distance by choosing Base station that is not shadowed (less deep fade) when compare to others the mobile unit can acquire a better signal to noise ratio in its forward path. Such contacting reduces large scale fading.

- **Types of Diversity:**

1. Space Diversity Technique
2. Polarization Diversity
3. Frequency diversity
4. Time Diversity

1. **Space Diversity:** This is most conventional method in which we have several antennas which are separated by different frequency band and they reach to receiver. Even if one got fade rest will reach to receiver. There are 4 types of subdivision:

- a. Selection Diversity
- b. Feedback Diversity
- c. Maximal Diversity
- d. Equal Gain Diversity

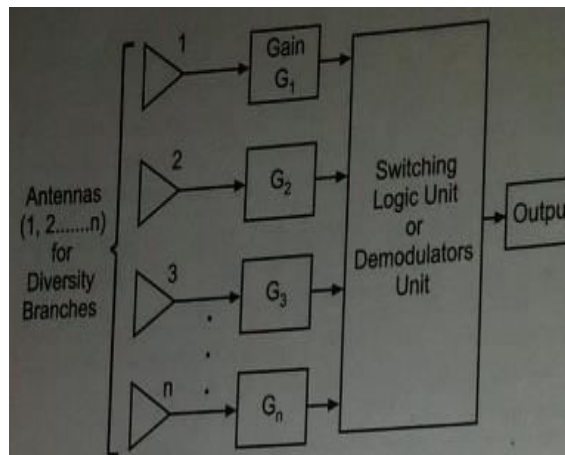


Fig. 3.22. Simple block diagram of space diversity technique

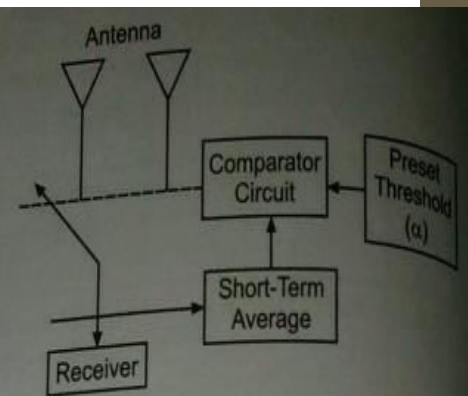


Fig. 3.23. Feedback diversity technique

2. Polarization Diversity: This is used where signal is transmitted using pair of polarized antenna and receiver by another pair of antennas then fading channel can be receiver. Hence this diversity is used. Circular and linear antenna are used.

3. Frequency Diversity: In this information is transferred/carried by many carrier frequency. Info is sent on carrier from transmitter end to receiver end.

4. Time Frequency: In time diversity information is transmitted repeatedly after a specific time. It leads to repetition of signal also.